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Understanding Logistic Regression: A Beginner's Guide

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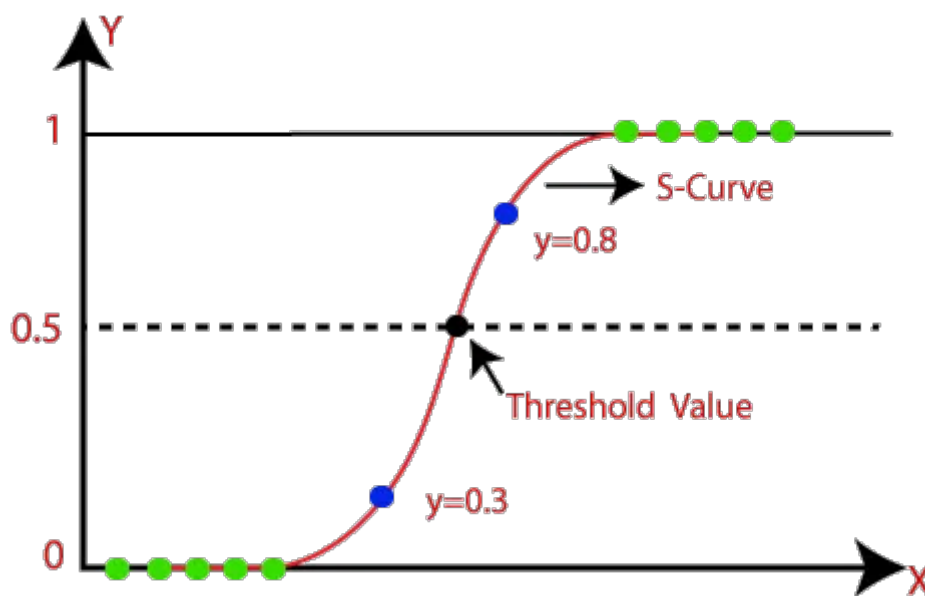


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Logistic Regression (image courtesy: javatpoint.com)

What is Logistic Regression?

Logistic regression is a supervised machine learning algorithm mainly used for classification tasks where the goal is to predict the probability that an instance belongs to a given class or not.

It is a type of statistical analysis that helps us understand the relationship between a set of input variables and an output variable that can take on **one of two possible values** (e.g. yes/no, true/false, 1/0). It does this by estimating the probability that the output variable will take on a certain value given the values of the input variables.

For example, logistic regression could be used to predict whether an email is spam or not based on certain characteristics of the email such as the sender, the subject line, and the content.

Why is it called regression if it is a classification algorithm?

The term “regression” in logistic regression refers to the fact that the model is trying to predict the probability of a categorical outcome, such as whether or not someone will buy a product. This is in contrast to linear regression, which is used to predict a continuous outcome, such as the price of a product.

In logistic regression, the probability of the outcome is represented by a logistic function. The logistic function is a sigmoid curve that has a range of values from 0 to 1. This means that the predicted probability of the outcome can be any value between 0 and 1.

The hypothesis of Logistic Regression

The hypothesis of Logistic Regression is based on the logistic function (also known as the sigmoid function). In logistic regression, we aim to model the probability of a binary outcome (e.g., 0 or 1) based on input features. The hypothesis function of logistic regression is defined as follows:

$$p(y = 1) = 1 / (1 + \exp(-(\beta_0 + \beta_1 x_1 + \beta_2 x_2)))$$

where:

- $p(y = 1)$ is the probability of the binary outcome being equal to 1
- β_0 is the intercept
- β_1 and β_2 are the coefficients for the predictor variables x_1 and x_2
- ε is the error term

The sigmoid function is a mathematical function that has an S-shaped curve. It is used to map the predicted values from the logistic regression model to a probability scale.

The logistic regression model is fit to the data using a maximum likelihood estimation (MLE) procedure. The MLE procedure is a statistical method that finds the parameters of a model that maximize the likelihood of the data.

Once the logistic regression model has been fit to the data, it can be used to predict the probability of a binary outcome for new data. The predicted probability is calculated as the output of the model, which is a linear combination of the predictor variables, with the sigmoid function applied to the result.

To convert the predicted probabilities into actual binary predictions, a threshold is often applied. For example, if the threshold is set at 0.5, any predicted probability above 0.5 is classified as the positive class ($y = 1$), while probabilities below 0.5 are classified as the negative class ($y = 0$).

Assumptions of Logistic Regression

Logistic Regression has the following assumptions:

1. The dependent variable must be **binary**. This means that the dependent variable can only take on one of two values, such as “yes” or “no”, “success” or “failure”, or “alive” or “dead”.
2. The independent variables must be linearly related to the log odds of the dependent variable. This means that the independent variables must have a linear relationship with the log odds of the dependent variable. The log odds is a measure of the odds of the dependent variable occurring, and it is calculated as the logarithm of the odds.
3. The errors must be **independent**. This means that the errors in the model must not be **correlated** with each other.
4. The errors must have a **constant variance**. This means that the errors in the model must have the same variance.
5. There must be **no multicollinearity** among the independent variables. This means that the independent variables must not be highly correlated with each

other.

Properties of Logistic Regression

Logistic regression has the following properties:

1. It is a linear model, meaning it is a model that is linear in the parameters.
2. It is a parametric model, meaning it is a model that is specified in terms of a set of parameters.
3. The parameters of logistic regression can be estimated using maximum likelihood estimation.
4. The predictions from logistic regression are probabilities that can be interpreted as the likelihood of the output variable taking on a certain value given the values of the input variables.
5. The logistic function used in logistic regression ensures that the predicted probabilities are always between 0 and 1.

These properties make logistic regression a powerful and flexible tool for classification tasks where the output variable can take on one of two possible values.

Advantages of Logistic Regression

Logistic Regression has the following advantages:

1. It is easy to implement and very efficient to train.
2. It provides a measure of how relevant an independent variable is for predicting dependent variable.
3. It can handle both continuous and categorical input variables.
4. The output of logistic regression is a probability that can be interpreted as the

likelihood of the output variable taking on a certain value given the values of the input variables.

5. Logistic regression is widely used and well understood, with many resources available for learning and troubleshooting.

These advantages make logistic regression a popular choice for classification tasks where the output variable can take on one of two possible values.

Limitations of Logistic Regression

Logistic Regression has the following limitations:

1. It requires that each data point be independent of all other data points.
2. It requires that there be little or no multicollinearity among independent variables.
3. It can only predict a categorical outcome.
4. Logistic regression assumes a linear relationship between the logit transformation of the dependent variable and the independent variables.

If this assumption is not met, the results may not be accurate. These limitations should be taken into account when using logistic regression for classification tasks.

Data Preparation for Logistic Regression

To implement logistic regression to a dataset, the data needs to be prepared, some of the important steps for data preparation are:

1. Clean and preprocess the data, handling missing values and outliers.
2. Select relevant features for the logistic regression model.
3. Encode categorical variables using techniques like one-hot encoding.

4. Scale continuous variables for better convergence during optimization.
5. Address class imbalance if present, using techniques like oversampling or undersampling.
6. Split the data into training and testing sets for model evaluation.
7. Check for multicollinearity and handle it if present.

Evaluation Metrics for Logistic Regression model

There are many evaluation metrics that can be used to assess the performance of a logistic regression model.

Some of the most common metrics include:

1. Accuracy
2. Precision
3. Recall
4. F1-score
5. ROC-AUC Curve

These evaluation metrics provide different perspectives on the model's performance and are used based on the specific requirements and goals of the problem at hand.

It is important to note that the evaluation metrics listed above are all for binary classification problems. For multi-class classification problems, there are a variety of other evaluation metrics that can be used.

Confusion Matrix

The confusion matrix is a tabular representation that summarises the performance of a classification model. It provides a detailed breakdown of the predicted and actual class labels. The confusion matrix for a binary classification problem consists of four cells:

		Predicted	
		1	0
Actual	1	TP	FN
	0	FP	TN

Confusion matrix

- **True Positive (TP):** The model correctly predicted the positive class.
- **True Negative (TN):** The model correctly predicted the negative class.
- **False Positive (FP):** The model incorrectly predicted the positive class when the actual class was negative (Type I error).
- **False Negative (FN):** The model incorrectly predicted the negative class when the actual class was positive (Type II error).

The confusion matrix helps evaluate the model's performance by calculating various metrics:

1. **Accuracy:** The overall proportion of correct predictions, calculated as $(TP + TN) / (TP + TN + FP + FN)$.
2. **Precision:** The proportion of true positive predictions among all positive predictions, calculated as $TP / (TP + FP)$.
3. **Recall (Sensitivity or True Positive Rate):** The proportion of true positive predictions among all actual positive instances, calculated as $TP / (TP + FN)$.
4. **Specificity (True Negative Rate):** The proportion of true negative predictions among all actual negative instances, calculated as $TN / (TN + FP)$.
5. **F1 Score:** The harmonic mean of precision and recall, providing a balanced measure of model performance, calculated as $2 * (precision * recall) / (precision + recall)$.

Conclusion

In conclusion, logistic regression is a powerful and widely used machine learning algorithm for classification tasks. It is easy to implement and efficient to train, and can handle both continuous and categorical input variables.

However, it also has some limitations and assumptions that must be taken into account when using it for classification tasks. By properly preparing the data and carefully evaluating the model's performance using appropriate metrics, logistic regression can be a valuable tool for solving real-world problems.

I hope this article has provided a useful introduction to logistic regression and its applications.

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$$RSS = \sum_{i=1}^n \left(y_i - \left(\beta_0 + \sum_{j=1}^p \beta_j x_{ij} \right) \right)^2$$

n is the total number of observations

y_i is actual output value of the observation

p is total number of features.

β_j is model's coefficient

x_{ij} is the i^{th} observation, j^{th} feature's value

p



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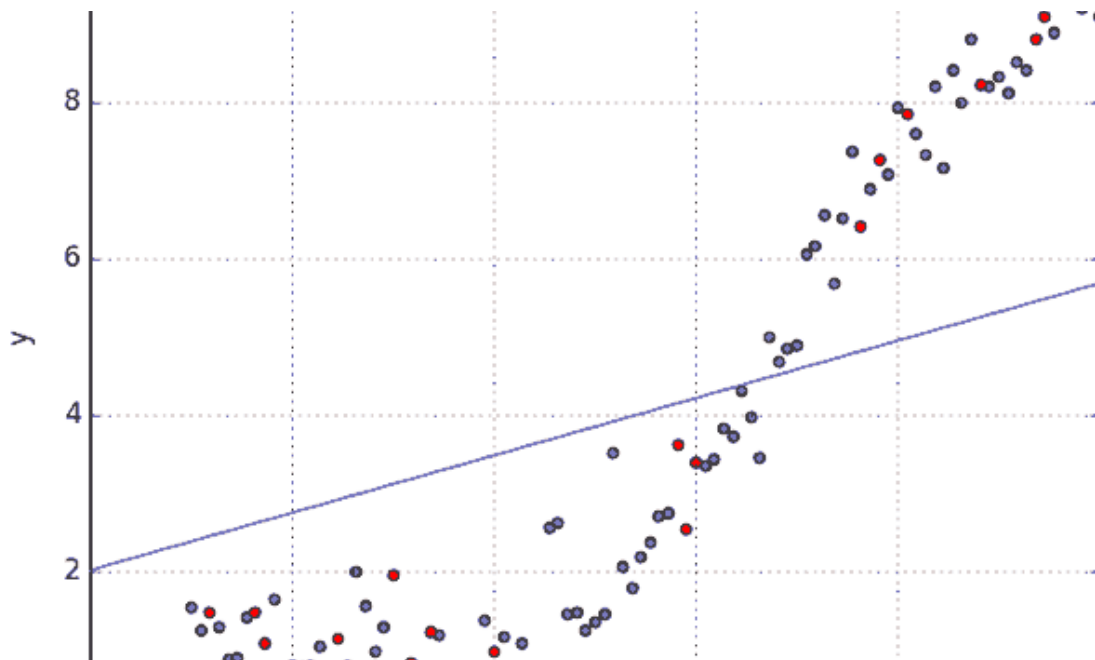
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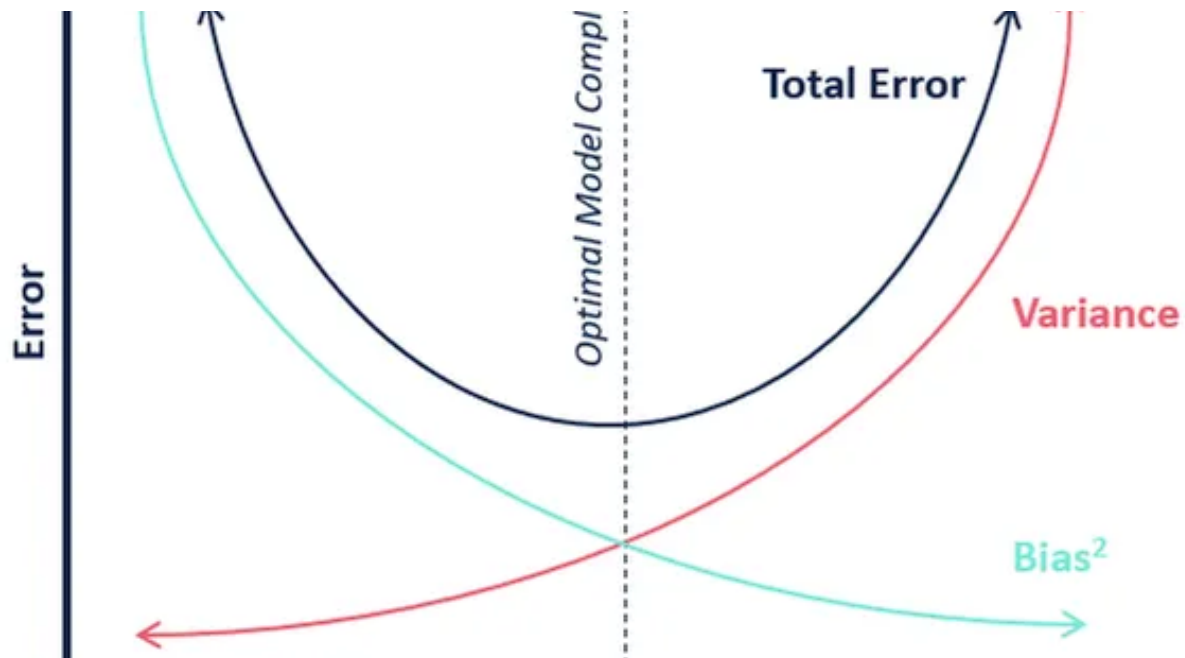
Linear Regression (An Overview)

What is Linear Regression?

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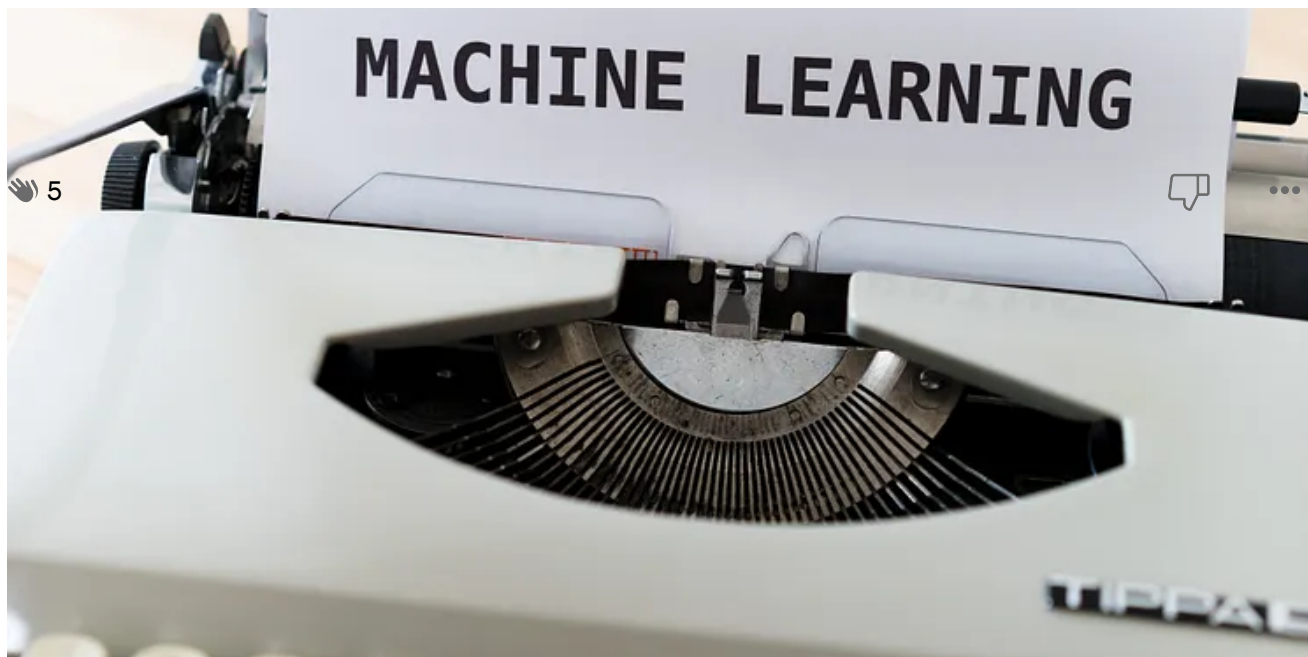


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